

Computer Vision Levels in a Real-Time Surveillance System

Question: A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.

Concept

Computer Vision (CV) is commonly divided into three levels:

- 1. Low-Level Vision** – Performs image enhancement, filtering, edge detection, and noise removal.
- 2. Mid-Level Vision** – Extracts meaningful information such as motion detection, segmentation, object tracking, and background subtraction.
- 3. High-Level Vision** – Understands scene content through object recognition, activity recognition, face recognition, and decision-making.

Classification

Answer: The given surveillance system belongs to **Mid-Level Computer Vision**.

Reason:

- It performs **background subtraction** to separate moving objects from the static background.
- It detects **motion** rather than identifying object categories.
- Motion detection is an intermediate processing task between image enhancement and scene understanding.
- This makes it suitable for real-time surveillance applications.

Code Explanation

1. Creates a black background image.
2. Generates a surveillance frame containing a white square (moving object).
3. Initializes the MOG2 Background Subtractor.
4. Learns the background and extracts the moving foreground.
5. Displays the detected motion mask.

Python Code

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

bg = np.zeros((100, 100), dtype=np.uint8)

frame = bg.copy()
cv2.rectangle(frame, (40, 40), (60, 60), 255, -1)

backSub = cv2.createBackgroundSubtractorMOG2()

backSub.apply(bg)
motion_mask = backSub.apply(frame)

plt.imshow(motion_mask, cmap='gray')
plt.title('Mid-Level CV: Extracted Motion Mask')
plt.show()
```

Result

The motion mask highlights the moving square while suppressing the static background. This demonstrates real-time motion detection using a Mid-Level Computer Vision technique.